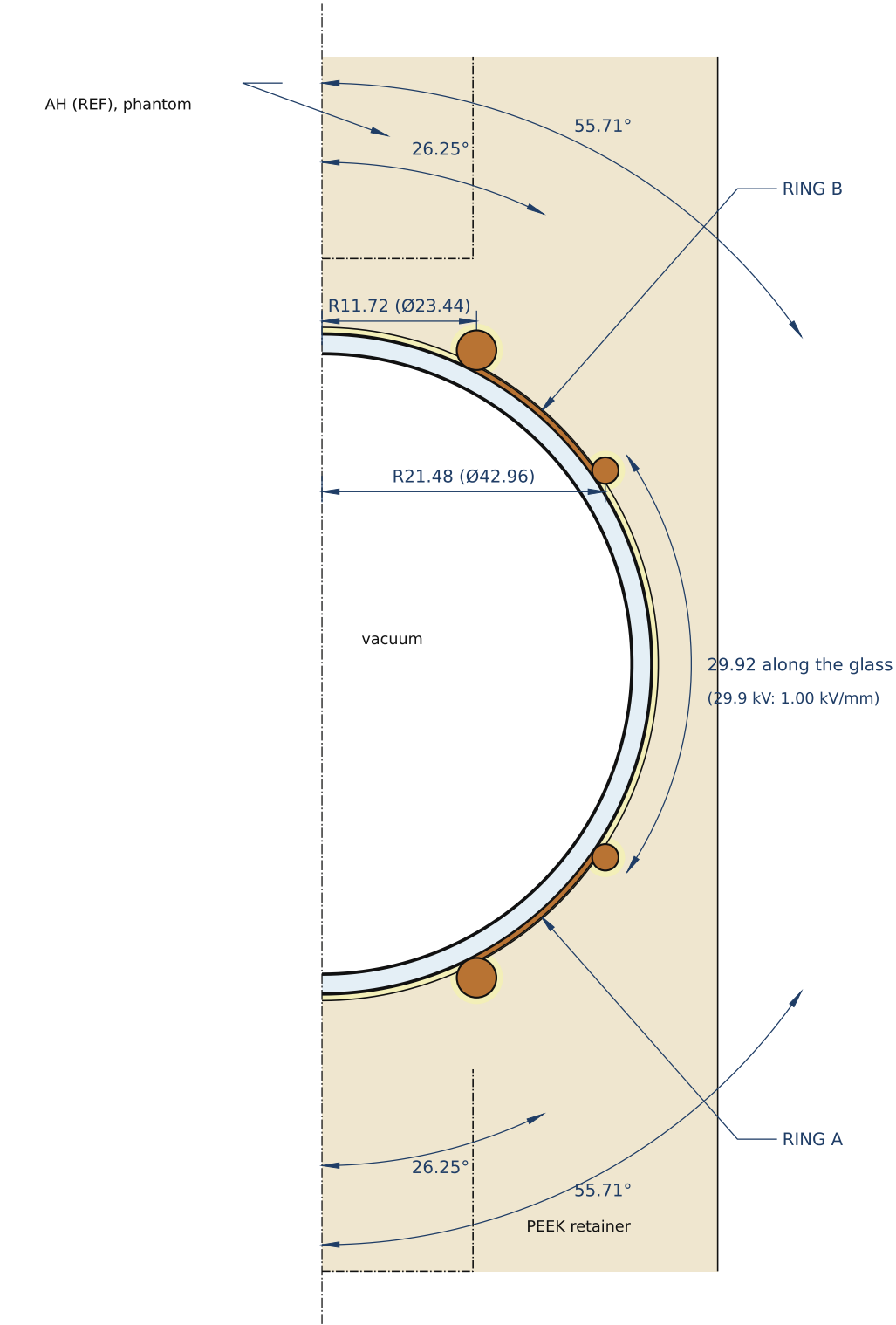
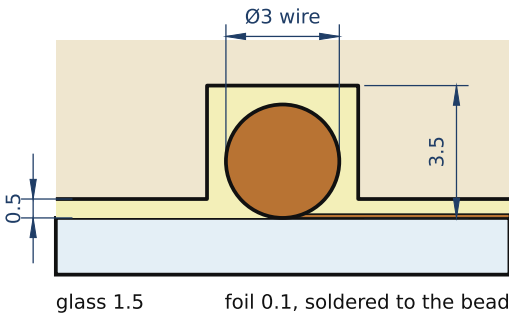


A HALF-SECTION ON THE AXIS (2:1)

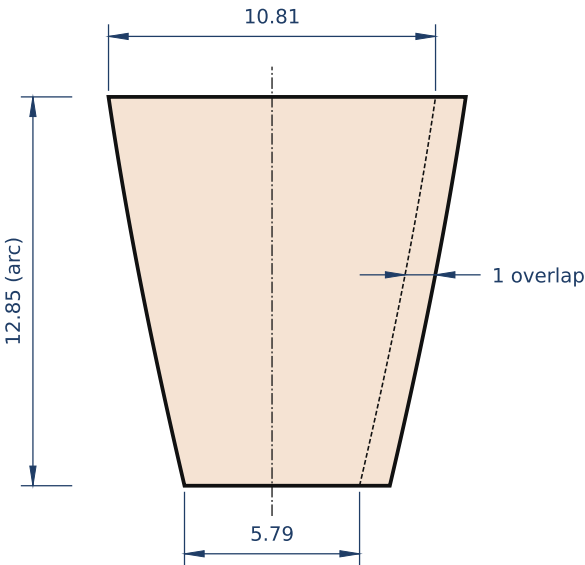
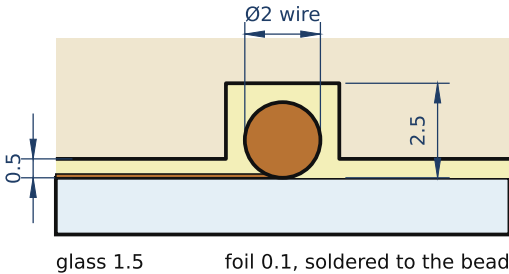
rings A and B mirror the equator; angles from the axis



B POLAR BEAD, Ø3 (5:1)



C EQUATORIAL BEAD, Ø2 (5:1)



D GORE, FLAT (4:1): 12 per band, 24 in all

width = $2\pi \cdot 25 \cdot \sin \theta / 12$ along the arc; the edge θ at the narrow end

RINGS A AND B: band 12.85 along the glass, 13.1 cm² each; the gap 29.92

feature (each ring)	polar edge	equatorial edge
edge angle from the axis	26.25°	55.71°
bead's contact on the glass	r 11.06 / z ±22.42	r 20.65 / z ±14.08
bead's centre	r 11.72 / z ±23.77	r 21.48 / z ±14.65
bead: Cu wire, ring centre-line	Ø3 wire, Ø23.44	Ø2 wire, Ø42.96
length per ring, the centre line (note 3)	73.6	135.0
groove in the PEEK (depth)	3.5	2.5
peak field in the gel (record)	4.85 kV/mm	4.36 kV/mm

NOTES

1. Material: foil Cu-ETP (CW004A), annealed, 0.10 thick; bead rings Cu-ETP wire. Glass: the vessel as supplied (borosilicate, OD 50, wall 1.5), clean and degreased.
2. Cut 12 gores per band (D), deburr every edge; burnish them into a gel film (no adhesive) from the polar edge, each overlapping its neighbour by 1; fillet each lap's free edge with solder, ≥ 0.2 wide. No void under a lap.
3. Polar rings (Ø3) seamless: turned, or brazed and turned round. Equatorial (Ø2): butt joint dressed round within 0.03 over 1. Solder each along its foil's edge at the edge angle (A), touching the glass; no solder ball, no burr.
4. The leads: PTFE-insulated HV wire soldered to the equatorial bead, out through the PEEK and the G10 coupler.
5. The PEEK retainer: unfilled, annealed stock; the pocket 0.5 over the glass and the grooves (B, C) machined to size; degassed silicone gel vacuum-cast into the pocket and the grooves, void-free, above all at the contacts.
6. Ring A is ring B mirrored in the equator. Test before potting: continuity; after: hold-off per docs/bench-test-rings.md phase 1 (c).

DCCREG turbine · the hub: the field's rings on the vessel

RINGS A AND B, COPPER, BEADED

Drawing no.

Cu-ETP foil 0.10 + wire; in gel under PEEK

DCCREG-HUB-201

Scale: 2:1 (A), 5:1 (B, C), 4:1 (D) Units: mm

Rev A (draft) Sheet 1/1 A3

Drawn: generated from sim/hub_rings_build.py

Date: 2026-10-09